

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q18-1124

Lagoon Creek 100 kV
January 23, 2020

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1124. The Interconnection Customer proposes a 237.5 MW solar photovoltaic facility interconnecting at the Lagoon Creek 100 kV substation in the Ohio area. The request was received on December 18, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 435 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 5 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$15,557,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
ALIANCE CASTINGS – SIMMONS 100kV	CRANBROOK – RAVENNA 100kV	159.3	97	8.3
OLIVEHURST – LAGOON CRK 100kV	LAGOON CRK – EIGHTEENTH STREET 100kV	146.1	102	41.2
VINE ST – BILLINGS EASTSIDE 100kV	ATASCOCITA – N CAMBRIDGE 230kV	122.8	115	5.4
OLIVEHURST – EPHRATA 100kV L	LAGOON CREEK – EIGHTEENTH STREET SUBSTATION 100kV	117.6	117	41.2
NEW MELONES – ARMSTRONG 345kV	FORT GIBSON – ARMSTRONG 500kV	108.5	1,061	5.1

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
ALLIANCE CASTINGS – SIMMONS SUBSTATION 100kV	Rebuild line with higher-capacity conductor	\$7,991,000	39% (\$3,119,000)
OLIVEHURST – LAGOON CRK 100kV	Rebuild line with higher-capacity conductor	\$3,774,000	100% (\$3,774,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
VINE ST – BILLINGS EASTSIDE 100kV	Rebuild line with higher-capacity conductor	\$3,110,000	16% (\$490,000)
OLIVEHURST – EPHRATA 100kV L	Rebuild line with higher-capacity conductor	\$7,556,000	100% (\$7,556,000)
NEW MELONES – ARMSTRONG 345kV	Reconductor line	\$15,084,000	4% (\$618,000)
	Total allocated to project		\$15,557,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.